

2023 秋概率统计标准答案和评分标准

一、 1)A, 2)D, 3)C, 4)A, 5)B

二、

1)4/7=0.571; 2)0.081; 3)1; 4)12; 5)0.4; 6)1; 7) $\frac{\pi^2}{2} - 2 = 2.935$; 8)1/40=0.025

$$9)F(x) = \begin{cases} 0 & x < 1 \\ 0.25 & 1 \leq x < 2 \\ 0.8 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10)f_Y(y) = \begin{cases} \frac{1}{4} & -1 < y < 3 \\ 0 & \text{其它} \end{cases}$$

11)1; 12)[13.76, 16.24]; 13)0.735

三、 1)

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \dots\dots\dots 1'$$

$$a \int_0^1 \int_0^x (x - y) dx dy = 1 \dots\dots\dots 2'$$

$$a = 6 \dots\dots\dots 2'$$

$$2)f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \dots\dots\dots 1'$$

$$\text{当 } x < 0, \text{ or } x > 1 \text{ 时, } f_X(x) = 0; \dots\dots\dots 1'$$

$$\text{当 } 0 < x < 1 \text{ 时,}$$

$$f_X(x) = \int_0^x 6(x - y) dy = 3x^2 \dots\dots\dots 3'$$

$$3)P(Y < 0.75 | X < 0.5) = \frac{P(Y < 0.75, X < 0.5)}{P(X < 0.5)} \dots\dots\dots 2'$$

$$P(X < 0.5, Y < 0.75) = \int_0^{0.5} \int_y^{0.75} 6(x - y) dx dy = 0.406 \dots\dots\dots 1'$$

$$P(X < 0.5) = \int_0^{0.75} 3x^2 dx = 0.75^3 = 0.422, \dots\dots\dots 1$$

$$P(Y < 0.75 | X < 0.5) = 0.406/0.422$$

$$= 0.962 \dots\dots\dots 1'$$

四、

B – 取出红球

A_1 – 取自甲箱; A_2 – 取自乙箱; A_3 – 取自丙箱

$$P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}; \dots\dots\dots 1'$$

$$P(B|A_1) = \frac{2}{3}; P(B|A_2) = \frac{2}{3}; P(B|A_3) = \frac{1}{2} \dots\dots\dots 1'$$

$$(1) P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \dots\dots\dots 2'$$

$$= \frac{11}{18} = 0.611 \dots\dots\dots 2'$$

$$(2) P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(B)} \dots\dots\dots 2'$$

$$= \frac{\frac{1}{3} \times \frac{2}{3}}{\frac{11}{18}} = \frac{4}{11} \approx 0.364 \dots\dots\dots 2'$$

五

	N	-1	0	1
M				
0		0	0.4	0
1		0.3	0	0.2
2		0	0.1	0

$$\dots\dots\dots 6'$$

$$P(M = 0) = 0.4; P(N = 0) = 0.5; \dots\dots\dots 2'$$

$$P(M = 0, N = 0) = 0.4 \neq P(M = 0) \times P(N = 0) = 0.2 \dots\dots\dots 1'$$

M 和 N 不相互独立。 $\dots\dots\dots 1'$

六、

$$\mu = EX = \int_0^1 x 3x^2 dx = \frac{3}{4}; EX^2 = \int_0^1 x^2 3x^2 dx = \frac{3}{5}$$

$$\sigma^2 = EX^2 - (EX)^2 = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{80}; \dots\dots\dots 4'$$

$X_1, \dots, X_n, n = 100$, 表示该样本

所求概率为

$$P(\sum_{i=1}^n X_i \leq 85) \approx \Phi\left(\frac{85 - n\mu}{\sqrt{n\sigma^2}}\right) \dots\dots\dots 4'$$

$$= \Phi\left(\sqrt{\frac{80}{3}}\right) \approx 1 \dots\dots\dots 1'$$

七、总体的分布律为: $f(x, p) = p(1-p)^{x-1}, x = 1, 2, \dots$ 1'

$$\begin{aligned}
 (1) \text{ 似然函数为: } L(p) &= \prod_{i=1}^n f(X_i, p) = \prod_{i=1}^n p(1-p)^{X_i-1} \\
 &= p^n (1-p)^{\sum X_i - n} \dots \dots \dots 1' \\
 \ln L(\theta) &= n \ln p + (\sum X_i - n) \ln(1-p) \dots \dots \dots 1' \\
 [\ln L(p)]' &= \frac{n}{p} - \frac{1}{1-p} (\sum X_i - n) = 0 \dots \dots \dots 1' \\
 \hat{p} &= \frac{1}{\bar{X}}; \quad \bar{X} = \frac{1}{n} \sum X_i \dots \dots \dots 1' \\
 \hat{\theta} &= \frac{1}{\hat{p}} = \bar{X} \dots \dots \dots 1'
 \end{aligned}$$

$$2) E\hat{\theta} = E\bar{X} = EX = \frac{1}{p} = \theta. \dots \dots \dots 3'$$

$\hat{\theta}$ 是 θ 的无偏估计。 $\dots \dots \dots 1'$

八、

$$(1) \text{ 检验统计量: } T = \frac{\bar{X} - \mu_0}{S_n} \sqrt{n} \sim t(n-1) (H_0 \text{ 成立时})$$

$$\text{拒绝域: } D = \left\{ |T| > t_{\frac{\alpha}{2}}(n-1) \right\} \dots \dots \dots 3'$$

$$n = 25, \mu_0 = 4, \bar{X} = 5, S_n = 2, \alpha = 0.05, t_{\frac{\alpha}{2}}(n-1) = 2.064;$$

$$T \text{ 的观测值: } T = \frac{5-4}{2} \sqrt{25} = 2.5 \dots \dots \dots 1'$$

$$|T| = 2.5 > 2.064 \dots \dots \dots 1'$$

所以, 拒绝原假设。 $\dots \dots \dots 1'$

$$2) \sigma^2 \text{ 的置信区间: } \left[\frac{(n-1)S_n^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}, \frac{(n-1)S_n^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)} \right] \dots \dots \dots 2'$$

$$\alpha = 0.05; S_n^2 = 4, \chi_{\frac{\alpha}{2}}^2(n-1) = 39.36, \chi_{1-\frac{\alpha}{2}}^2(n-1) = 12.4; \dots \dots \dots 1'$$

所求置信区间为: $[2.44, 7.74] \dots \dots \dots 1'$